

A general framework for learning interpretable dynamical laws directly from cell trajectories.

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I. INTRODUCTION

Over decades, substantial effort has been devoted to identifying the forces and mechanisms that drive cell dynamics in diverse environments [1–3]. Cells move under the combined influence of cell–cell interactions, external chemical and mechanical cues, intrinsic active driving, and stochastic fluctuations [4–6]. Disentangling these contributions remains challenging, and is the focus of this work.

Recent advances in imaging and segmentation have generated a wealth of high-resolution cell tracking data, both *in vitro* and *in vivo*. Cell positions are typically extracted by tracking nuclei via fluorescent markers, digital staining, or by computing centroids from Voronoi tessellations of cell–cell junctions [7–9]. These developments have enabled quantitative studies of collective migration across a wide range of biological contexts, providing new insights into coordinated cell motion during embryonic development [10], epithelial wound healing [11], and migration in engineered microenvironments [12].

Separately, there has been rapid progress in using machine learning to infer force laws and equations of motion directly from particle trajectories, in cases where the functional form of the interactions is not known *a priori* [13]. Graph neural network architectures are particularly well-suited to this problem, as they naturally encode pairwise interactions between particles while respecting physical symmetries [14–16]. Applied to stochastic trajectories, such approaches can recover interaction potentials without assuming their form [17], and have been extended to active matter systems including self-propelled colloids [18], yielding interpretable physical parameters. Together, these developments establish a toolkit for learning the physics of complex many-body systems from observations alone.

ML tools have also recently been applied directly to collective cell migration. Using trajectory data alone, deep attention networks can learn cell-type-specific migration rules and reveal which neighbouring cells most influence a given cell’s motion [19]. Graph neural networks have shown that collective migratory dynamics can be predicted even from a static snapshot of cell positions, and that cell geometry, junction networks, and cell–cell interactions are key predictors of single-cell behaviour [20–23]. Together, these approaches demonstrate that cell dynamics are more predictable than previously thought. Extending this toolkit towards interpretable, physics-based descriptions of the underlying forces and timescales remains an open goal.

A key challenge in applying force inference to cells is

that cell trajectories exhibit persistent, correlated motion arising from internal polarity and cytoskeletal dynamics [24, 25]. While polarity can be measured in specialised experiments and therefore modelled explicitly [26], the vast majority of cell tracking data consists only of nuclear or centroid positions, leaving these internal degrees of freedom unobserved. Standard inference frameworks are not equipped to handle such hidden dynamical variables. In constrained geometries, this has been approximated by modelling cells with a second-order equation of motion, successfully revealing cell–cell interactions [27] and three-body effects [28]. A generalised inference scheme for freely migrating cells that explicitly accounts for these hidden dynamics is lacking.

Here we develop a physics-informed graph neural network inference framework and apply it to freely migrating cells, a setting in which such interpretable force-inference tools have not previously been used. By explicitly modelling hidden persistent dynamics, the framework recovers not only cell–cell interaction forces but also a characteristic persistence timescale for each cell type, a quantity inaccessible to prior cell force-inference methods. We first validate the framework on simulated particles in nonlinear external force fields, demonstrating accurate recovery of both the force landscape and memory timescale. Next, we apply the framework to experimental *E. coli* trajectories in micropillar confinement, benchmarking against criteria derived from the known geometry of the environment. We then extend the framework to crowded many-body systems. Finally, we apply the framework to experimental MDCK cell-tracking data to characterise dynamical state transitions between epithelial-like and mesenchymal-like phenotypes.

II. SYSTEM AND INFERENCE SCHEME

We present an inference framework that explicitly models hidden memory while retaining an interpretable dynamical description that yields physically meaningful quantities, such as characteristic persistence times.

A. System

As shown in Fig. 1, migrating cells often have a ‘memory’ of their dynamics, shown by correlations in their velocity over some characteristic timescale τ . For single migrating cells in a homogeneous environment, this memory often decays exponentially, suggesting that the cells migrate according to an overdamped equation of motion,

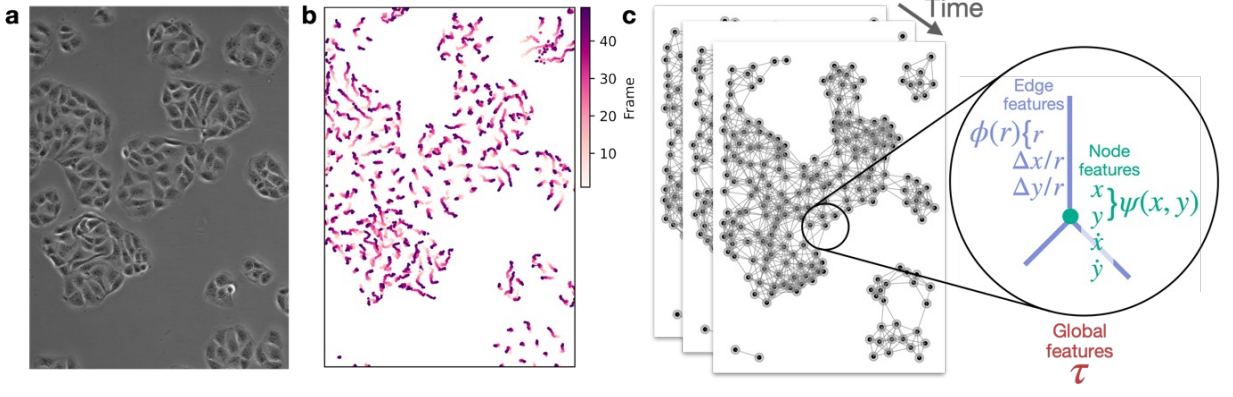


FIG. 1. **Cell dynamics and inference scheme.** **a)** Phase contrast snapshot of MDCK cells. **b)** Individual trajectories over 50 frames (1 frame=10 minutes) of the same MDCK cells as in **a)**. **c)** Inference scheme of this work. Three consecutive frames of the cells are shown in graph representation with node and edge features indicated. Edge features, node features and global features all contribute to the predicted equations of motion.

where each cell is propelled by its own active persistent term $\mathbf{f}^A$ [25, 29].

$$\gamma \dot{\mathbf{x}}_i = \mathbf{F}_i + \mathbf{f}_i^A \quad (1)$$

$$\dot{\mathbf{f}}_i^A = -\frac{1}{\tau} \mathbf{f}_i^A + \boldsymbol{\xi} \quad (2)$$

where $\mathbf{x}_i$ is cell i 's position, $\boldsymbol{\xi}$ is Gaussian white noise with zero mean and correlations $\langle \xi_\alpha(t) \xi_\beta(t') \rangle = \sigma^2 \delta_{\alpha\beta} \delta(t - t')$, γ is the drag a cell experiences from its environment, and bold font indicates d -dimensional vectors, with all our results being in 2D.

Here τ is defined as the relaxation timescale of the unresolved active propulsion variable $\mathbf{f}^A$, and σ sets the amplitude of the fluctuations that drive it. Here τ is defined as the relaxation timescale of the unresolved active-propulsion variable f^A . In the simulations results, where f^A is explicitly generated by Eq. (2), τ is the known intrinsic persistence timescale. For experimental trajectories, Eq. (2) should be regarded as an effective coarse-grained description of unobserved persistent propulsion, and the inferred τ is correspondingly an effective persistence timescale of those hidden dynamics. All other forces, including cell-cell interactions, a chemical field or hard boundaries, are for now represented by the force $\mathbf{F}_i$. We aim to create an inference scheme which is successful at recovering the forces and memory timescales when we have incomplete information, i.e. we do not have access to the direction of the self-propulsive term $\mathbf{f}^A$. Such incomplete information is typical of most cell-tracking experiments. Following [30], we differentiate Eq. (1) with respect to time and substitute in Eq. (2), to obtain:

$$\tau \gamma \ddot{\mathbf{x}}_i = \mathbf{F}_i + (\tau \nabla_{\mathbf{x}} \mathbf{F}_i - \gamma \mathbf{I}) \cdot \dot{\mathbf{x}}_i + \tau \boldsymbol{\xi} \quad (3)$$

Here $\nabla_{\mathbf{x}} \mathbf{F}_i$ is the Jacobian of $\mathbf{F}_i$ with respect to the spatial coordinates, with components $(\nabla_{\mathbf{x}} \mathbf{F}_i)_{\alpha\beta} = \partial F_{i,\alpha} / \partial x_{i,\beta}$, and $\mathbf{I}$ is the $d \times d$ identity matrix. Throughout, $\cdot$ denotes contraction of the second index of a matrix with a vector, $(\mathbf{A} \cdot \dot{\mathbf{x}})_\alpha = \sum_\beta A_{\alpha\beta} \dot{x}_{i,\beta}$, so the bracketed term is a $d \times d$ matrix acting on the velocity rather than a scalar prefactor. For now we assume that $\mathbf{F}_i$ depends explicitly only on $\mathbf{x}$.

$$\dot{\mathbf{x}}_i = \mathbf{v}_i \quad (4)$$

$$\dot{\mathbf{v}}_i = \mathbf{F}_i + \gamma_{\text{eff}} \cdot \mathbf{v}_i + \boldsymbol{\xi} \quad (5)$$

Where we have rescaled $\mathbf{F}_i \rightarrow \mathbf{F}_i / (\tau \gamma)$ and $\boldsymbol{\xi} \rightarrow \boldsymbol{\xi} / \gamma$. Eq. (5) encompasses the basic model we assume going forward for cell motility. The important difference between this model and a purely underdamped equation of motion is that the effective drag, $\gamma_{\text{eff}} = (\tau \nabla_{\mathbf{x}} \mathbf{F}_i - \frac{1}{\tau} \mathbf{I})$ is now space-dependent and is a function of the interpretable memory timescale, τ [30]. This is the assumed model for the inference in sections III A & III B.

B. Extension to pairwise interactions

The force $\mathbf{F}_i$ on a cell may not only depend on its position in space but on other cells which surround it. In this case the total force on cell i splits as $\mathbf{F}_i = \mathbf{F}_i^{\text{env}} + \mathbf{F}_i^{\text{int}}$, where $\mathbf{F}_i^{\text{env}}(\mathbf{x}_i)$ is the single-cell environmental (space-dependent) force and $\mathbf{F}_i^{\text{int}} = \sum_{j \neq i} \mathbf{f}_{ij}^{\text{int}} = \sum_{j \neq i} f(r_{ij}) \hat{\mathbf{r}}_{ij}$ is the pairwise interaction force summed over all other cells j , with $r_{ij} = |\mathbf{x}_i - \mathbf{x}_j|$ and $\hat{\mathbf{r}}_{ij} = (\mathbf{x}_i - \mathbf{x}_j) / r_{ij}$.

The step to obtain an equivalent Eq.(5) from Eqs. (1)&(2) in this case requires the additional time derivative of the interaction term, as also shown previously [31, 32]. Differentiating $\mathbf{F}_i^{\text{int}}$ with respect to time

introduces velocities through $\dot{\mathbf{f}}_{ij}^{\text{int}} = \mathbf{M}_{ij} \Delta \mathbf{v}_{ij}$, where $\Delta \mathbf{v}_{ij} = \dot{\mathbf{x}}_i - \dot{\mathbf{x}}_j$ and

$$\mathbf{M}_{ij} = \frac{df}{dr} \hat{\mathbf{r}}_{ij} \hat{\mathbf{r}}_{ij}^\top + \frac{f(r_{ij})}{r_{ij}} (\mathbf{I} - \hat{\mathbf{r}}_{ij} \hat{\mathbf{r}}_{ij}^\top). \quad (6)$$

This encodes the radial (along $\hat{\mathbf{r}}_{ij}$) and transverse components of the change in force due to cell-cell relative motion. Note that the i, j indices are labeling particle pairs and not matrix/vector indices. A derivation of matrix $\mathbf{M}_{ij}$ in (6) is outlined in Appendix A 1. $\mathbf{I}$ is the identity matrix. All other bold font quantities are vectors with spatial components.

Since $\Delta \mathbf{v}_{ij} = \dot{\mathbf{x}}_i - \dot{\mathbf{x}}_j$, the pairwise memory contribution splits into a term linear in $\dot{\mathbf{x}}_i$ and a term linear in $\dot{\mathbf{x}}_j$. We separate the terms linear in $\dot{\mathbf{x}}_i$ and define a new effective drag, with the terms linear in $\dot{\mathbf{x}}_j$ contributing to a new effective force, $\mathbf{H}_i$:

$$\gamma'_{\text{eff},i} = \tau \nabla_{\mathbf{x}} \mathbf{F}_i^{\text{env}} + \tau \sum_{j \neq i} \mathbf{M}_{ij} - \frac{1}{\tau} \mathbf{I}, \quad (7)$$

$$\mathbf{H}_i = -\tau \sum_{j \neq i} \mathbf{M}_{ij} \dot{\mathbf{x}}_j, \quad (8)$$

Where $\gamma'_{\text{eff},i}$ is now a 2x2 matrix. Equation (5) therefore generalises to

$$\dot{\mathbf{v}}_i = \mathbf{F}_i^{\text{env}} + \mathbf{F}_i^{\text{int}} + \mathbf{H}_i + \gamma'_{\text{eff},i} \cdot \mathbf{v}_i + \boldsymbol{\xi}. \quad (9)$$

with the re-defined effective drag $\gamma'_{\text{eff},i}$ now containing spatially varying pairwise contributions of the form $\tau f(r)/r$ and $\tau f'(r)$, and $\mathbf{H}_i$ treated as an additional force term in inference. Versions of equation (9) have been widely used as an effective description of systems with persistent dynamics, from active matter exhibiting motility-induced phase separation [33] to nonequilibrium models of associative memory [34]. This is the assumed model for sections III C & III D.

C. Inference scheme

In our inference scheme we aim to take trajectories which are derived from time-lapse microscopy images of cells, see 1A&B, and infer the persistent memory of the cells, τ , as well as all the forces governing their motion, $\mathbf{F}^{\text{env}}$ and $\mathbf{F}^{\text{int}}$. We start by setting up the trajectory information in the form of graphs, with one graph for each frame. Each node is a cell, with features such as its velocity and position, and edges connect two cells, with features such as cell-cell distance, shown in Fig. 1C.

The functional form of the forces, $\mathbf{F}^{\text{env}}$ and $\mathbf{F}^{\text{int}}$, in Eq. (9) are *a priori* unknown, and we therefore would like them to be represented as flexibly as possible. For this reason, we represent the forces as Multi-Layer Perceptrons (MLPs) which can take the form of some arbitrary non-linear function. Specifically, the neural network representing the environment term depends on the

position, (x, y) of each cell and outputs a vector that directly contributes to the forces: $\psi^{\text{env}} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with learnable parameters being the weights of the MLP. This may represent a global chemical field which is position-dependent and which all cells experience in the same way. The scalar interaction network $\phi^{\text{int}} : \mathbb{R} \rightarrow \mathbb{R}$ takes a normalised interparticle distance as input and returns a force magnitude. The interaction net outputs are then multiplied by the unit vectors connecting node edges to produce the force vector, see Fig. 1C. This represents cell-cell interactions and we assume these interactions are additive, reciprocal and radial. Lastly, we have the learnable global scalar parameter τ . The final predicted force, which we distinguish from the true force using $\hat{\cdot}$ symbol, for a particular cell is the sum of the inferred forces: $\hat{\mathbf{F}}_i = \psi_i^{\text{env}} + \sum_{i \neq j} \phi_i^{\text{int}} + \mathbf{H}(\phi_i^{\text{int}}, \tau) + \gamma'_{\text{eff},i}(\tau) \cdot \mathbf{v}_i$. Note that $H(\phi_i^{\text{int}}, \tau)$ and $\gamma'_{\text{eff},i}(\tau) \cdot \mathbf{v}_i$ are *calculated* from the velocities and pairwise distances of pairs of particles (not trained), see Eqs. 8&7, and contain just one learnable scalar, τ .

We define a loss function:

$$\mathcal{L} = \langle (\dot{\mathbf{v}}_i - \hat{\mathbf{F}}_i)^2 \rangle + 2K\sigma^2 \langle \nabla_{\mathbf{v}_i} \cdot \hat{\mathbf{F}}_i \rangle \quad (10)$$

where the averages are across nodes, i , frames and dimensions. The multiplicative factor K is a scalar and depends on the derivative stencils we use, see SI section VIC. The first term is the mean squared error between the true dynamics $\dot{\mathbf{v}}$ and our representation of the dynamics $\hat{\mathbf{F}}_i$. The second term is derived from finding the bias from discrete estimates of our acceleration and velocity resulting from stochastic dynamics, following [35–37]. We show the full derivation in the SI VIC.

Note that the forces inferred are all re-scaled from real units by a factor $1/\gamma\tau$ where γ is the (*not* effective) drag relating forces to velocities in Eq. (1). We optimize parameters for ϕ^{int} , ψ^{env} and τ using an Adam optimizer to minimize this loss for a given dataset. The noise amplitude σ cannot itself be optimised against Eq. (10), which is linear in σ^2 . It is instead estimated from the data via the residual of the fit, $\hat{\sigma}^2 \propto (\Delta t) \langle (\dot{\mathbf{v}}_i - \hat{\mathbf{F}}_i)^2 \rangle$, with the proportionality constant dependent on the derivative stencils used (SI VIC). Full details of the ML model and training details are given in SI VIA 3

D. Influence of measurement noise

Experimental measurements inevitably contain noise: positions are discretised to pixel resolution, and centroid localisation introduces uncertainty when cells deform, overlap, or move rapidly between frames [38, 39]. Even modest noise levels can severely bias inference. Accelerations are estimated as second finite differences of position which amplifies noise in a manner which is not readily accounted for in the model. An effective way to handle this is to avoid naïve finite differences and instead compute derivatives from local windows of the data [40–42].

Here we apply a polynomial smoothing method called the Savitzky-Golay (S-G) filter [41, 43, 44]. Velocity and accelerations are then computed directly from the polynomial stencil coefficients of the filter. This approach recovers correct inferred forces even when measurement noise is large enough to make naïve derivatives fail catastrophically (SI Fig. 6). In the interacting-particle case, for instance, even small measurement noise causes the inferred persistence timescale to be consistently underestimated without filtering (see SI Fig. 6d). See SI Sec. VID for further details on the measurement noise assumptions.

A less obvious consequence of filtering concerns the bias-correction factor which must be applied to the effective drag, mentioned in the above section and derived in Appendix A 4-5. With naïve finite differences this factor is $\frac{2}{3}$ [35–37], arising from the overlap structure of the derivative stencils. Switching to an S-G filter changes the stencil overlap, and hence the correction factor, which must be re-derived accordingly. We show this calculation in SI Section A 3. The resulting scalar correction factor depends only on the filter window size and polynomial degree. All experimental results in Sections IIIB and IIID use S-G filtered trajectories with this correction applied.

Also add note about additional correction factor to loss from measurement error estimate

III. RESULTS

A. Recovery of forces and timescales from simulated trajectories in external fields

We first validate the hidden-dynamics inference framework in a setting where the ground-truth dynamics are known. We simulate particles undergoing persistent random motion in a nonlinear external force field. Nonlinear forces generate position-dependent gradients, meaning effective drag of the second order dynamics (Eq. (5)) is spatially dependent. This provides a simple test of whether the inference pipeline (Fig. 1C) can simultaneously recover both the environmental forces and the hidden memory timescale.

We simulate Eqs. (1) and (2) with a two-dimensional nonlinear trap of the form $F_{x/y} = -\kappa \text{sign}(x/y) |(x/y)|^\alpha$, illustrated in Fig. 2A. Unless otherwise stated, simulations use $\tau = 10 T_{\text{sim}}$, $\gamma = 1$, $L = 1$, and $\kappa_x = \kappa_y = 0.5 k_B T / \sigma^2$. Other simulation details are in Appendix VIB. An example trajectory is shown in Fig. 2B.

We apply our hidden-variable inference framework to jointly infer the force field and memory timescale. We compare two inference schemes: the full overdamped-with-memory formulation, Eq. (5), which allows for a spatially varying effective drag, and a naïve underdamped scheme that assumes a constant effective drag. We quantify model performance on the inferred force via $R_F^2 = 1 - \frac{\sum (F^T - F^I)^2}{\sum (F^T - \bar{F}^T)^2}$, where F^T is the true non-linear

force, $\bar{F}^T$ is the mean of the non-linear force over the data and F^I is the model predictions. The summations are over the data positions and summed over x, y . $R_F^2 = 1$ for perfectly inferred forces, $R_F^2 = 0$ when a model is only able to capture the mean and $R_F^2 < 0$ for a model worse than the mean. When we vary the nonlinearity parameter α we find that the two approaches perform similarly at $\alpha = 1$ (Fig. 2C). This is expected: a linear force ($\alpha = 1$) produces a constant force gradient, so the effective drag is spatially uniform and the constant-drag assumption of the underdamped scheme is correct in this regime. As soon as $\alpha > 1$, however, the force gradient becomes position-dependent and the two schemes begin to deviate, with the overdamped-with-memory formulation outperforming the underdamped approximation across the remainder of the range explored. Examining the inferred force field directly for the default parameters, the overdamped-with-memory formulation recovers a landscape with an error relative to the intrinsic noise in the dynamics that is everywhere $\ll 1$ (Fig. 2Di). In contrast, using the same trajectory data, the underdamped scheme produces an error-to-noise ratio approaching order unity across much of the domain (Fig. 2Dii). When re-simulating the inferred force fields for $\alpha = 1.5$ and $\kappa = 1 k_B T$, the overdamped-with-memory model ($R_F^2 = 0.96$) reproduces all six measured collective statistics within the seed-to-seed variability of the original simulation, while the underdamped model ($R_F^2 = 0.87$) falls outside this range for 4/5 metrics (Fig. 8). This demonstrates that a sufficiently accurate overdamped-with-memory force field is not merely a good fit to training data, but faithfully recovers the statistical collective behaviour of the original system.

In force-free systems, the persistence timescale is often estimated from the velocity autocorrelation function (VACF). However, the external force field modifies the velocity dynamics and biases this measurement. As shown in Fig. 2Ei, the VACF decays much faster than the theoretical prediction for free persistent motion (derivation in appendix A 2), leading to an apparent persistence time that is shorter than the true input value. The external field therefore obscures the intrinsic active memory. By contrast, our hidden-dynamics framework accurately recovers the input memory timescale across the full range of α explored (Fig. 2Eii).

Beyond the nonlinearity parameter α , we examine how performance and the inferred timescale depend on the other simulation parameters: the force strength κ and the memory timescale τ (Fig. 7). As expected, the forces are inferred to a higher accuracy when the trap strength is larger (Fig. 7E), and the signal to noise ratio is higher (Fig. 7A&E). The performance of the underdamped model degrades further for larger memory timescales, where the true space-dependent drag term becomes more important (Fig. 7D&F). The overdamped-with-memory inference is always able to infer the correct timescale when the forces are inferred accurately ($R^2 > 0.85$), see Fig. 7B&E. Together, these results

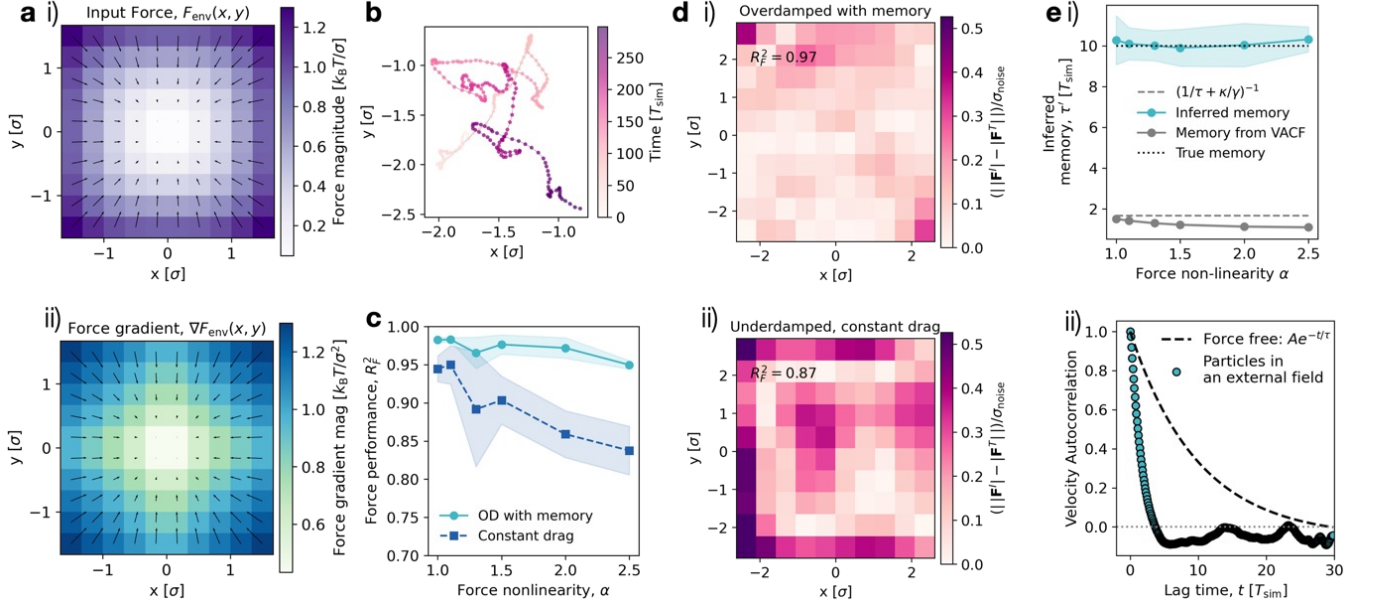


FIG. 2. **Non-linear external force field.** **a)** Particles move in a non-linear external field, where the force gradient is dependent on the spatial coordinate. **b)** Example of a simulated trajectory over 300 frames (colour indicates time). **c)** The performance of the inferred model, $R_F^2 = 1 - \frac{\sum (F^T - F^I)^2}{\sum (F^T - F^T)^2}$ as the non-linearity parameter α is varied. **d)** The error on the inferred external field. The error is significantly smaller for the correct inference scheme with a non-constant effective drag, Eqs. (5) (i), compared to a naïve scheme using underdamped dynamics with a constant drag (ii). **e) i)** The velocity autocorrelation of the particles deviates from that expected if the particles were following Eqs. (1)&(2) with no external forces. **e) ii)** The inferred memory timescale as the non-linearity parameter α is varied.

demonstrate that explicitly accounting for hidden persistence dynamics is essential for correctly inferring forces and memory timescales in spatially varying force landscapes, where standard VACF-based approaches and simple underdamped approximations fail.

B. The framework identifies physically consistent force fields in confined bacterial trajectories

We next apply our inference framework to experimental trajectory data from bacteria migrating through structured microfluidic environments containing arrays of micropillars [45]. In these experiments, *E. coli* move through environments with varying levels of confinement, producing trajectories shaped by both intrinsic run-and-tumble dynamics and steric interactions with the pillar walls. Representative trajectories are shown in Fig. 3a. For these experimental trajectories we treat Eq. (2) as an effective description of the unresolved propulsion dynamics and the inferred τ therefore quantifies their effective persistence rather than being assumed a priori to represent an environment-independent intrinsic bacterial timescale. Since the micropillar geometries are spatially periodic, trajectories are folded onto a reduced domain consisting of four quarter-pillars, substantially increasing local sampling density and making force inference tractable (Fig. 3b). Cell-cell interactions are negligible

given the low bacterial density in these experiments. We compare the overdamped-with-memory inference to the underdamped formulation with constant drag, evaluating each against three physically motivated criteria.

First, we assess how well each model reproduces the marginal distribution of true accelerations via the Kullback-Leibler (KL) divergence,

$$D_{\text{KL}}(p_{\text{data}} \| p_{\theta}) = \int p_{\text{data}} \log \frac{p_{\text{data}}}{p_{\theta}} d\mathbf{v}, \quad (11)$$

where p_{data} and p_{θ} are estimated from samples via Kernel Density Estimation (KDE) over a shared grid in acceleration space. $D_{\text{KL}} = 0$ indicates a perfect match and larger values reflect poorer overlap. The overdamped-with-memory model achieves consistently lower KL divergence than the underdamped model (Fig. 3e).

Second, a physically well-grounded inference should produce large force gradients near the pillar walls, where bacteria are most strongly redirected by the geometry, and small gradients in the open channels away from the walls. The magnitude of the spatial gradient of the inferred force, $|\nabla|\mathbf{F}_{\text{env}}||$, is shown for both models in Figs. 3c,d. The overdamped-with-memory model recovers gradients that are clearly concentrated at the pillar rims (Fig. 3c), in line with physical expectation. The underdamped model, by contrast, fails to produce pillar-consistent force gradients.

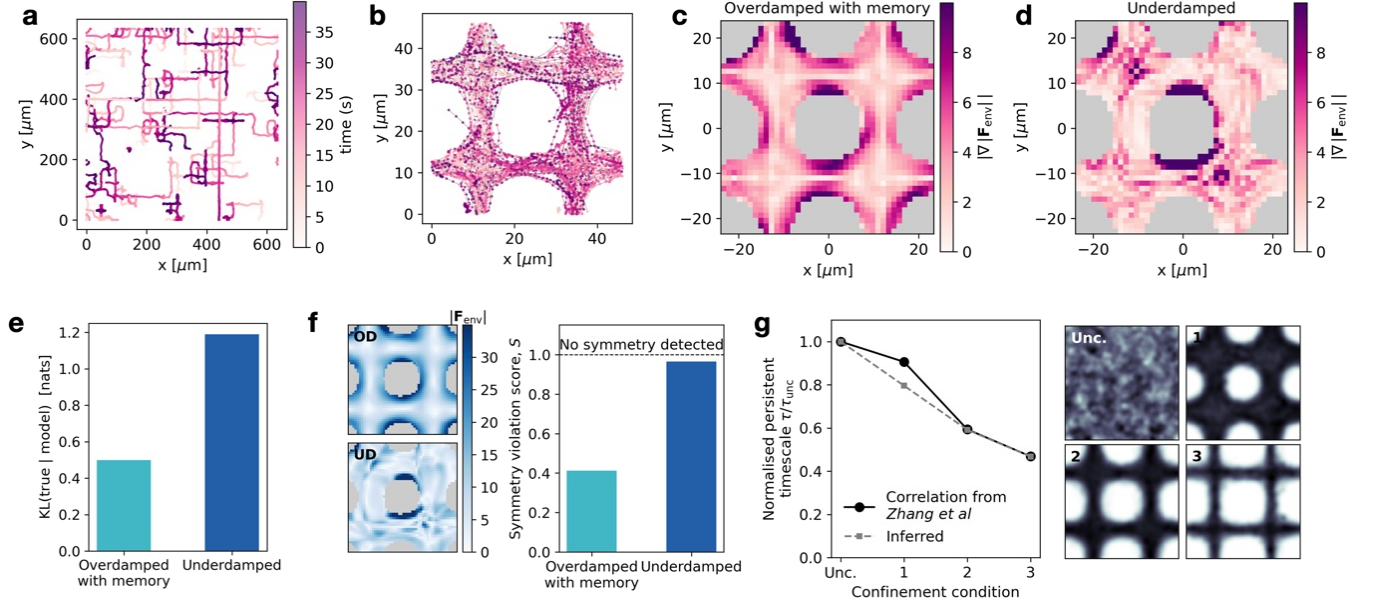


FIG. 3. **External force field for confined bacterial motion.** **a)** Trajectories of bacteria migrating through a micropillar confinement geometry, coloured by time, adapted from [45]. **b)** To exploit the spatial symmetry of the system and improve sampling density, trajectories are folded onto a reduced unit-cell domain consisting of four quarter-pillars. Coloured by the same time as in **a)**. **c,d)** Gradient of the inferred spatial force field $|\nabla|\mathbf{F}_{\text{env}}||$ for the overdamped-with-memory (**c**) and underdamped (**d**) inference frameworks, evaluated on the folded domain. Grey regions indicate low data density. **e)** Inference performance quantified by the KL divergence between true and predicted acceleration distributions; lower values indicate better agreement. The overdamped-with-memory model substantially outperforms the naïve underdamped model. **f)** Reflection symmetry violation score S for each model (left: inferred force magnitude; right: bar chart), where $S = 0$ indicates perfect symmetry and $S = 1$ indicates maximum asymmetry. The overdamped-with-memory model recovers the expected fourfold symmetry of the pillar geometry, while the underdamped model does not. **g)** Persistence timescale τ normalised to the unconfined value τ_{unc} across confinement conditions (Unc., 1, 2, 3), estimated via velocity autocorrelation function (VACF) fitting and via the hidden-dynamics inference framework. Insets show representative folded density maps for each condition.

Lastly, the square pillar lattice imposes fourfold reflection symmetry on the environmental force field, which a correct inference should respect. For reflections in x , we define two symmetries: $S_x^x = F_x(x, y) + F_x(-x, y)$ and $S_x^y = F_x(x, y) - F_x(x, -y)$ and we have the equivalent in y -symmetries. We then quantify violations using a symmetry score.

$$S = \sqrt{\frac{\langle (S_x^x)^2 + (S_x^y)^2 + (S_y^x)^2 + (S_y^y)^2 \rangle}{4\langle |\mathbf{F}|^2 \rangle}} \quad (12)$$

Here $S = 0$ indicates perfect symmetry and $S = 1$ indicates that violations are as large as a random field. The overdamped-with-memory model recovers a nearly symmetric force field, while the underdamped model approaches maximum asymmetry (Fig. 3f). Taken together, all three criteria consistently identify the overdamped-with-memory formulation as the physically consistent inference.

Finally, we compare persistence timescales inferred using our framework with those obtained from correlation-based measurements (Fig. 3g), where the correlation time is defined as the time when the measured VACF reaches 10% of its initial value, as calculated in [45]. For the

unconfined condition, the average VACF decays exponentially with a correlation time of 1.4s, consistent with the assumed dynamics Eq. (2). We infer an effective persistence timescale of $\tau = 1$ s for the unconfined trajectories. Interestingly, we find that the inferred timescale decreases with increasing confinement, tracking the VACF correlation time closely. This suggests that wall encounters reset the self-propulsion direction. Such reorientation is known to happen both passively, because the cell body hydrodynamically reorients and drags the flagellar thrust with it [46], and actively, when the bacterium tumbles upon wall contact [47]. Since the inter-pillar spacing is much shorter than the free-swimming run length of *E. coli* [45], wall encounters dominate the reorientation statistics, setting the effective τ .

We verified that all reported results, including the inferred force field and the inferred timescale, are robust to the choice of folded domain size (number of quarter-pillars used) and to the MLP architecture (activation function, number of layers, and number of neurons per layer). A limitation of this analysis is that the pillar walls are physically hard boundaries, yet here they are represented by a smooth, continuous force field parametrised by an MLP. This approximation may fail to capture the

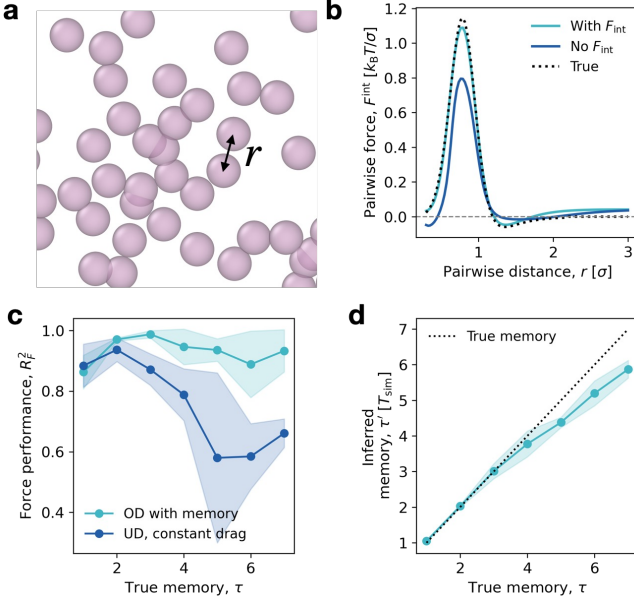


FIG. 4. **Pairwise interactions in a crowded persistent system.** **a)** Snapshot of the simulated multi-particle system, in which 100 persistent particles interact via a soft Lennard–Jones potential in a periodic box. **b)** Inferred pairwise interaction force as a function of pairwise distance r for the overdamped-with-memory (teal) and underdamped constant-drag (dark blue) inference schemes, compared to the true force (dotted), shown here for $\tau = 3.0 T_{\text{sim}}$. **c)** Force inference performance $R_F^2 = 1 - \frac{\sum (F^I - F^T)^2}{\sum (F^T - \bar{F}^T)^2}$ as a function of true memory timescale τ for the overdamped-with-memory (teal) and underdamped constant-drag (dark blue) schemes, with shaded regions indicating the standard deviation across three random seeds. **d)** Inferred memory timescale $\hat{\tau}$ versus true memory timescale τ , with the identity line (dotted) indicating perfect recovery

sharp force gradients at the wall surface, and could introduce artefacts in the inferred dynamics close to the pillar rims.

C. Recovery of pairwise interactions and persistence timescales in crowded simulated systems

We next consider a crowded multi-particle environment with pairwise interactions, mimicking a tissue-like setting in which cell–cell interactions dominate the dynamics. In such systems, persistent motion is largely masked by collisions and confinement, making it difficult to infer intrinsic persistence directly from trajectories. We extend the hidden-dynamics inference framework to this many-body setting, jointly inferring pairwise interaction forces and the intrinsic persistence timescale from multi-particle trajectory data.

We simulate $N = 100$ particles in a two-dimensional periodic box of side length $L = 15\sigma$, with packing fraction set such that inter-particle interactions are frequent.

Particles interact via a soft-core Lennard–Jones potential with interaction strength ε , length scale $\sigma = 1.1$, softness parameter $\lambda = 0.1$, and cutoff $r_{\text{cut}} = 2\sigma + 1$. Unless otherwise stated, simulations use $\varepsilon = 0.3 k_B T$, $\gamma = 5$, $T = 0.1 k_B / \sigma^2$, and $dt = 0.001 T_{\text{sim}}$, with frames recorded every 100 steps. A simulation snapshot is shown in Fig. 4a, illustrating the dense, disordered configuration typical of these conditions.

We apply the hidden-dynamics inference framework to recover both the pairwise force profile and the memory timescale from the multi-particle trajectories. For a representative memory timescale $\tau = 7.0 T_{\text{sim}}$, the overdamped-with-memory formulation accurately recovers the shape and magnitude of the true pairwise interaction force across the full range of particle separations (Fig. 4b, teal). In contrast, the naïve underdamped inference scheme with a constant effective drag substantially misrepresents the force profile (Fig. 4b, dark blue).

Varying the true memory timescale τ and quantifying inference performance, we find that the overdamped-with-memory model maintains high performance in R_F^2 across the full range explored, with only a modest decline at the largest values of τ (Fig. 4c, teal). The underdamped approximation, by comparison, degrades substantially at longer memory timescales, reflecting its inability to separate inertial-like effective dynamics from genuine persistent activity. The inferred memory timescale tracks the true input value closely (Fig. 4d), with a slight systematic underestimation at large τ .

When re-simulating the inferred force fields for $\tau = 3.0 T_{\text{sim}}$ and $\epsilon = 0.3 k_B T$, the overdamped-with-memory model ($R_F^2 = 0.99$) reproduces all five measured collective statistics within the seed-to-seed variability of the original simulation (Fig. 10). The underdamped model ($R_F^2 = 0.84$) is worse than the overdamped on all measures, and its pair distribution function is not within seed-to-seed variability of the original, results that are summarised in Fig. 10d.

Together, these results show that the hidden-dynamics framework successfully disentangles pairwise interaction forces from intrinsic persistence even in dense, many-body settings, where neither trajectory-level autocorrelations nor simple underdamped approximations provide reliable estimates.

D. Characterisation of dynamical state transitions in experimental MDCK trajectories

TBC depends on which data.

IV. DISCUSSION

V. TO DO LIST/OUTSTANDING QUESTIONS

- τ is the solution to a quadratic equation so we often have two seemingly physical solutions. Only found

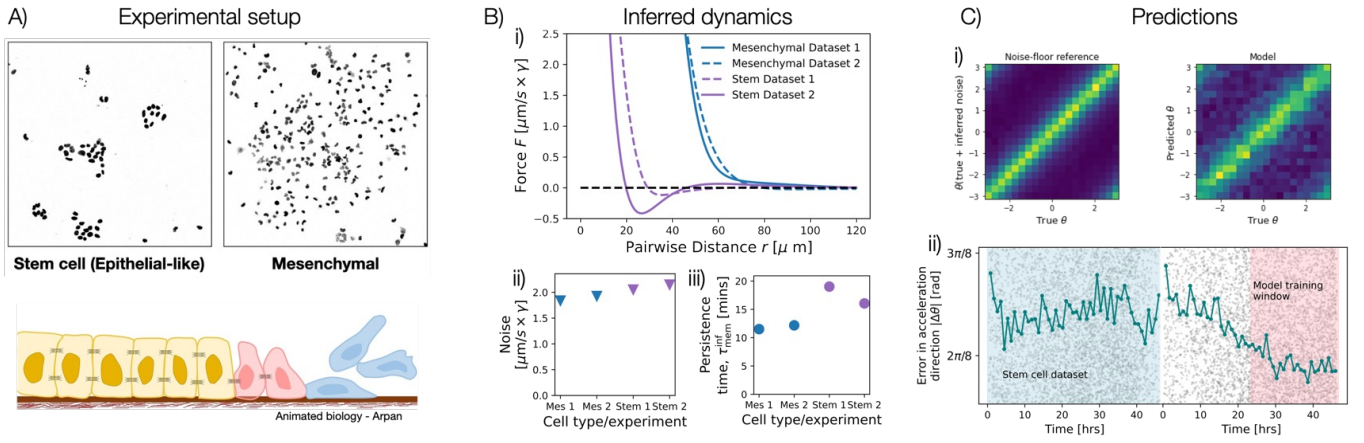


FIG. 5. **Experimental cell state transitions.**

a way to solve this by re-simulation, or comparison to force-free case. We need to put a comment about this.

- Remember to thank Adrienne, Federica, and Nico
- MDCK Figure 5
- Could: put all 3 confinement conditions in Fig3 e&f
- check architecture/training across datasets like the $\log(\tau)$ parametrisation and number of epochs and if we use far-field loss.

VI. SUPPLEMENTARY INFORMATION

A. Inference details

1. Data processing

a. From trajectories to graphs. One graph is constructed per frame. Each node is a cell or particle and carries its position and estimated velocity, $[x, y, \hat{v}_x, \hat{v}_y]$. The target is the estimated acceleration $[\hat{v}_x, \hat{v}_y]$. For the datasets with a pairwise term (the pairwise simulation and MDCK) edges connect every pair separated by less than a cutoff r_{cut} and carry the separation and unit vector $[r_{ij}, \hat{r}_{ij,x}, \hat{r}_{ij,y}]$. For the remaining datasets the interaction network is absent and no edges are required. Graphs are recomputed independently for every frame.

b. Derivative estimates and track preparation. Velocities and accelerations are estimated from the raw tracked positions. The scheme used for each dataset is listed in Table I: naïve finite differences for the simulations, and Savitzky–Golay derivatives with window $w = 2m + 1 = 9$, degree $d = 3$ and shift $S = 1$ for the experimental data. Both windows must fit inside a track, so only interior frames $t \in [m + S, n - 1 - m]$ of a track of length n yield a usable sample. Nodes with no usable derivative are masked, see below.

For the *E. coli* data the imaged region contains a periodic array of micropillars, and several unit cells are folded onto one to improve statistics. Derivatives are computed before folding, on the unfolded coordinates, so that the estimates remain accurate for cells whose derivative window straddles a domain boundary.

c. Node masking. Nodes for which no derivative estimate is available are retained in the graph, so that they still contribute to their neighbours’ interaction terms, but are excluded from the loss. For the MDCK data we additionally mask nodes within $20\mu\text{m}$ of the edge of the field of view, since those cells interact with cells outside the image and their predicted interaction force would be incomplete. No other dataset requires this. The fraction of nodes entering the loss is given in Table II.

d. Normalisation. All normalisation constants are computed once, by pooling every node and every edge over all frames of the dataset, and are then applied to every frame. Positions are centred component by component and scaled by a single isotropic factor, $x_\alpha \rightarrow (x_\alpha - \bar{x}_\alpha)/\sigma_\mathbf{x}$ with $\sigma_\mathbf{x} = (\sigma_x^2 + \sigma_y^2)^{1/2}$, where $\alpha \in \{x, y\}$ and $\bar{x}_\alpha$ and σ_α are the mean and standard deviation of component α . Pair separations are centred and scaled by their mean and standard deviation, $r_{ij} \rightarrow (r_{ij} - \bar{r})/\sigma_r$. The acceleration targets are scaled by their standard deviation $\hat{v}_\alpha \rightarrow \hat{v}_\alpha/\sigma_{\hat{v}}$.

e. Train/validation/test split. Frames are shuffled and assigned to training, validation and test sets in the proportions 70/15/15.

Dataset	Δt	Estimator	w	d	S	Φ
Simulated, external field (Sec. III A)	0.1 (LJ)	naïve	—	—	0	2/3
Simulated, pairwise (Sec. III C)	0.1 (LJ)	naïve	—	—	0	2/3
<i>E. coli</i> , micropillars (Sec. III B)	0.05 s	S-G	9	3	1	149/378
MDCK (Sec. III D)	10 min	S-G	9	3	1	149/378

TABLE I. Derivative estimation and the corresponding discretisation correction Φ (Appendix VIC). Naïve denotes the symmetric acceleration paired with the backward velocity at the same frame. For the experimental datasets the S-G window length w and degree d are selected by the residual-autocorrelation criterion of Fig. 6b.

Dataset	tracks	frames	nodes/frame	r_{cut}	nodes in loss
Simulated, external field	100	300	100	—	99.3%
Simulated, pairwise	100	300	100	3σ	99.3%
<i>E. coli</i> , micropillars	3143	800	~ 150	—	93.6%
MDCK	[FILL]	[FILL]	[FILL]	[FILL]	[FILL]%

TABLE II. Dataset sizes and graph construction. A dash for r_{cut} indicates that the dataset has no pairwise interaction term, so no edges are built. “Nodes in loss” is the fraction of nodes not masked.

2. Architecture

a. Implementation. The model is implemented in PyTorch 2.0.1. Savitzky–Golay stencil coefficients are obtained from `scipy.signal`. Unless stated otherwise, all remaining settings are library defaults, including weight initialisation.

b. The force networks. The environmental force is represented by a fully connected network $\psi^{\text{env}} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ taking a normalised position and returning a force vector. The pairwise term is represented by a scalar network $\phi^{\text{int}} : \mathbb{R} \rightarrow \mathbb{R}$ taking a normalised separation and returning a force magnitude, which is multiplied by the unit vector $\hat{r}_{ij}$ so that the interaction is radial, reciprocal and additive by construction. Widths, depths and activations are listed in Table III. Activations were chosen per dataset, but we obtained indistinguishable results with tanh, SiLU and ReLU activations.

c. Parameterisation of $\hat{\tau}$. The memory timescale is optimised as $\log \hat{\tau}$, so that $\hat{\tau} > 0$ by construction and the search is multiplicative rather than additive.

Dataset	ψ^{env}	ϕ^{int}	width	activation
Simulated, external field	3 hidden layers	—	64	tanh
Simulated, pairwise	—	3 hidden layers	64	tanh
<i>E. coli</i> , micropillars	5 hidden layers	—	64	SiLU
MDCK	—	[FILL]	[FILL]	[FILL]

TABLE III. Network architecture per dataset. ψ^{env} maps $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ and ϕ^{int} maps $\mathbb{R} \rightarrow \mathbb{R}$; a dash indicates the term is absent. In addition each model carries the single learnable scalar $\log \hat{\tau}$.

3. Training

The same procedure is used for every dataset. All parameters, including $\log \hat{\tau}$, are optimised jointly with Adam at a fixed learning rate of 10^{-3} , with one optimiser step per frame, for 200 epochs. No weight decay or other regularisation is applied. A far-field penalty on ϕ^{int} at large separations is available in the implementation but was not used for any result reported here. [FILL]- What did we do for MDCK?

This bit may have to be moved below section C-General stencil so it makes more sense. The correction term of Eq. (41) requires the dynamical noise amplitude σ^2 , which is not known *a priori* and we estimate it from the residual. Repeating the calculation of Eq. (28) for two acceleration stencils rather than an acceleration and a velocity stencil gives

$$\langle \delta \hat{v}_{i\mu} \delta \hat{v}_{j\nu} \rangle = \delta_{ij} \delta_{\mu\nu} \frac{\sigma^2 K_{aa}}{\Delta t}, \quad K_{aa} = \sum_{k,k'} c_k^a c_{k'}^a G(k, k'), \quad (13)$$

with G as in Eq. (22). The noise contribution to the measured acceleration therefore has variance $\sigma^2 K_{aa} / \Delta t$ per Cartesian component, and attributing the full residual of a converged fit to this contribution gives the estimator

$$\hat{\sigma}^2 = \frac{\Delta t}{K_{aa}} \left\langle \left(\hat{v}_{i\mu} - D_{\theta, i\mu} \right)^2 \right\rangle. \quad (14)$$

It is evaluated on the training data every 25 epochs.

B. Simulation details

Trajectories are generated by integrating Eqs. (1) and (2) directly, with custom code. Two systems are used: a non-interacting system in a non-linear external field, and an interacting system with pairwise forces and no external field. Parameters for both are collected in Table IV.

a. Integration scheme. Both systems are advanced by the Euler–Maruyama scheme. At each step the active force is updated first, $\mathbf{f}_i^A \leftarrow \mathbf{f}_i^A + (-\mathbf{f}_i^A / \tau + \boldsymbol{\xi}_i) dt$ with $\boldsymbol{\xi}_i \sim \mathcal{N}(0, 2T\gamma/dt)$, so that the increment of $\mathbf{f}^A$ has variance $2T\gamma dt$ per component. Positions are then displaced using the updated active force, $\mathbf{x}_i \leftarrow \mathbf{x}_i + (\mathbf{f}_i^A + \mathbf{F}_i^c) dt / \gamma$, where $\mathbf{F}^c$ is the confining or interaction force described below.

b. Non-linear external field, non-interacting. The environmental force acts independently on each Cartesian component, $F_{i,\mu}^{\text{env}} = -\kappa \text{sgn}(x_{i,\mu}) |x_{i,\mu}|^\alpha$ for $\mu \in \{x, y\}$, reducing to a linear (harmonic) trap at $\alpha = 1$ and increasingly non-linear with increasing α . The box is large enough that the trap confines every particle far from the soft repulsive walls, which are never reached during the simulations.

c. Pairwise interactions, periodic boundaries. Particles interact via a soft-core Lennard–Jones potential, $E(r) = 4\epsilon\lambda^n (A^{-2} - A^{-1})$ with $A = \alpha_{\text{LJ}}(1 - \lambda)^2 + (r/\sigma)^6$, truncated at r_{cut} , where $n = 1$ and $\alpha_{\text{LJ}} = 1/2$. The box is periodic, with pair separations taken under the minimum-image convention.

Parameter	Symbol	Non-linear field	Pairwise, PBC
Number of particles	N	100	100
Box size	L	100	15
Boundary condition	—	soft walls	periodic
Integration timestep	dt	0.001	0.001
Noise strength	T	0.1	0.1
Drag coefficient	γ	1	5
Memory timescale	τ	10	3
Confinement strength	κ	0.5	—
Field nonlinearity	α	1, 1.1, 1.3, 1.5, 2.0, 2.5	—
Wall interaction energy	ϵ_w	500	—
Particle diameter	σ	1.1	1.1
Interaction energy	ϵ	—	0.3
Core softening	λ	—	0.1
Pair cutoff	r_{cut}	—	$2\sigma + 1 = 3.2$
Equilibration steps	N_{eq}	40,000	40,000
Production steps	N_{prod}	30,000	30,000
Frame interval	Δt	$100 dt = 0.1$	$100 dt = 0.1$
Total frames	N_{frames}	300	300
Independent seeds	—	3	3

TABLE IV. Simulation parameters. Both systems are two-dimensional. A dash indicates that the term is absent from that system: the non-linear-field runs have no pairwise interaction, and the periodic runs have no external field.

C. General-stencil correction to the inference loss

The velocity and acceleration entering the loss are both estimated from the same stochastic position trajectory. They therefore contain contributions from the same underlying noise realization. These contributions are individually zero-mean, but their product need not average to zero. In underdamped inference this produces an $\mathcal{O}(\Delta t^0)$ bias in velocity-dependent terms, even as the sampling interval becomes small [35, 36].

Our goal in this section is therefore to identify this shared-noise correlation for the derivative estimators used here, and then determine how it shifts the fixed point of the mean-squared inference loss. We first carry out the calculation for arbitrary linear position stencils, so that the same result applies to both naïve finite differences and Savitzky–Golay (S-G) derivatives. We then use the resulting correlation to construct a modified loss whose leading discretisation bias cancels, without assuming conservative forces or $\langle \mathbf{v} \cdot \mathbf{F} \rangle = 0$. The analysis is closely related to Ref. [36], but is extended here to arbitrary velocity and acceleration stencils and incorporated directly into our inference through a modified loss.

For compactness we collect the complete deterministic right-hand side of the effective dynamics, Eq. (9), into

$$\mathbf{D}_i = \mathbf{F}_i^{\text{env}} + \mathbf{F}_i^{\text{int}} + \mathbf{H}_i + \gamma'_{\text{eff},i} \cdot \mathbf{v}_i, \quad (15)$$

for particle i , so that

$$d\mathbf{v}_i = \mathbf{D}_i dt + \sigma d\mathbf{W}_i. \quad (16)$$

Below, i and j label particles, while $\mu, \nu \in \{x, y\}$ label Cartesian components when component notation is required. We assume additive isotropic noise that is independent between particles and Cartesian components,

$$\langle dW_{i\mu} dW_{j\nu} \rangle = \delta_{ij} \delta_{\mu\nu} dt. \quad (17)$$

1. Shared noise in position-based derivative estimates

To determine the correlation between the stochastic parts of the measured velocity and acceleration, we first express both estimators in terms of short-time position increments. Let t_0 be the earliest time entering either derivative stencil and define

$$\Delta \mathbf{x}_i^n \equiv \mathbf{x}_i(t_0 + n\Delta t) - \mathbf{x}_i(t_0), \quad n \geq 0. \quad (18)$$

Following the short-time Itô–Taylor expansion, which can be found in Ref. [36],

$$\Delta \mathbf{x}_i^n = \mathbf{v}_i(n\Delta t) + \frac{1}{2} \mathbf{D}_i(n\Delta t)^2 + \sigma \mathbf{I}_i^n + \mathcal{O}(\Delta t^{5/2}), \quad (19)$$

where $\mathbf{v}_i$ and $\mathbf{D}_i$ are evaluated at t_0 , and

$$\mathbf{I}_i^n = \int_{t_0}^{t_0+n\Delta t} (t_0 + n\Delta t - s) d\mathbf{W}_i(s) \quad (20)$$

is the contribution of the dynamical white noise to the position increment.

The same $\mathbf{I}_i^n$ terms enter the position combinations used to estimate both velocity and acceleration. Their covariance therefore determines the leading shared-noise contribution to the measured acceleration–velocity correlation. Using the Itô isometry [48],

$$\langle I_{i\mu}^n I_{j\nu}^m \rangle = \delta_{ij} \delta_{\mu\nu} \Delta t^3 G(n, m), \quad (21)$$

with

$$G(n, m) = \frac{q^2(3p - q)}{6}, \quad q = \min(n, m), \quad p = \max(n, m). \quad (22)$$

We now write the measured acceleration and velocity as arbitrary linear combinations of the positions entering their respective stencils,

$$\hat{\mathbf{v}}_i = \frac{1}{\Delta t^2} \sum_k c_k^a \mathbf{x}_i(t_0 + k\Delta t), \quad (23)$$

$$\hat{\mathbf{v}}_i = \frac{1}{\Delta t} \sum_l c_l^v \mathbf{x}_i(t_0 + l\Delta t). \quad (24)$$

Here k and l are non-negative temporal offsets from the common earliest time t_0 . A relative shift of the velocity estimator by S frames is included simply by associating the velocity coefficients c_l^v with the appropriately shifted set of offsets l . For example, for the S-G scheme used here the acceleration window is centred one frame later than the velocity window.

For any valid first- or second-derivative stencil,

$$\sum_l c_l^v = 0, \quad \sum_k c_k^a = 0. \quad (25)$$

Therefore the common position $\mathbf{x}_i(t_0)$ cancels from Eqs. (23)–(24), and the estimators can equivalently be written in terms of the increments $\Delta \mathbf{x}_i^n$ of Eq. (18). Substituting Eq. (19), we denote by $\delta \hat{\mathbf{v}}_i$ and $\delta \hat{\mathbf{v}}_i$ the terms proportional to the Wiener increments:

$$\delta \hat{v}_{i\mu} = \frac{\sigma}{\Delta t^2} \sum_k c_k^a I_{i\mu}^k, \quad (26)$$

$$\delta \hat{v}_{i\mu} = \frac{\sigma}{\Delta t} \sum_l c_l^v I_{i\mu}^l. \quad (27)$$

The acceleration noise scales as $\Delta t^{-1/2}$ while the velocity noise scales as $\Delta t^{1/2}$. Their product is therefore $\mathcal{O}(\Delta t^0)$, so its average can remain finite as $\Delta t \rightarrow 0$. This is the origin of the discretisation bias.

Using Eq. (21),

$$\langle \delta \hat{v}_{i\mu} \delta \hat{v}_{j\nu} \rangle = \delta_{ij} \delta_{\mu\nu} \sigma^2 K_{av}, \quad (28)$$

where all dependence on the derivative scheme is contained in

$$K_{av} = \sum_{k,l} c_k^a c_l^v G(k, l). \quad (29)$$

Equation (28) is the quantity required for the inference correction: it measures the part of the apparent acceleration–velocity correlation generated purely because the two derivative estimates are constructed from the same stochastic position increments.

For the backward-velocity/symmetric-acceleration finite-difference scheme $K_{av} = 1/6$. For the S-G scheme used for the experimental data ($w = 9$, $d = 3$, $S = 1$),

$$K_{av} = \frac{229}{756}. \quad (30)$$

2. How the shared-noise correlation biases the loss

We next ask how Eq. (28) affects the parameters learned by the model. Let $\mathbf{D}_{\theta,i}$ denote the deterministic dynamics predicted by the model, with θ containing the MLP parameters and $\hat{\tau}$. We write the mean-squared loss as

$$\mathcal{L}_0 = \left\langle \left(\hat{v}_{i\mu} - D_{\theta,i\mu} \right)^2 \right\rangle, \quad (31)$$

The index $i \in \{1, 2, \dots, N\}$ denotes particle ID and $\mu \in \{x, y\}$ denotes Cartesian component. As in the main text, the average is over frames, particles and Cartesian components. If the measured velocity were noise-free, minimizing this loss would give the usual regression fixed point. However, $D_{\theta,i\mu}$ itself depends on the estimated velocities, so its stochastic error is correlated with that in measured $\hat{v}_{i\mu}$. Equation (28) quantifies this correlation.

For any learnable parameter θ_p ,

$$\frac{\partial \mathcal{L}_0}{\partial \theta_p} = -2 \left\langle \left(\hat{v}_{i\mu} - D_{\theta,i\mu} \right) \frac{\partial D_{\theta,i\mu}}{\partial \theta_p} \right\rangle. \quad (32)$$

Define

$$c_{i\mu,p} \equiv \frac{\partial D_{\theta,i\mu}}{\partial \theta_p}. \quad (33)$$

Because both $D_{\theta,i\mu}$ and $c_{i\mu,p}$ are evaluated using the estimated velocity, we first expand them about the true velocity. Writing $\hat{v}_{j\nu} = v_{j\nu} + \delta \hat{v}_{j\nu}$, where $\delta \hat{v}_{j\nu}$ denotes the stochastic contribution to the velocity estimator, we have to first order

$$c_{i\mu,p}(\hat{\mathbf{v}}) = c_{i\mu,p}(\mathbf{v}) + \sum_{j,\nu} \frac{\partial c_{i\mu,p}}{\partial v_{j\nu}} \delta \hat{v}_{j\nu} + \mathcal{O}(\delta \hat{v}^2), \quad (34)$$

$$D_{\theta,i\mu}(\hat{\mathbf{v}}) = D_{\theta,i\mu}(\mathbf{v}) + \sum_{j,\nu} \frac{\partial D_{\theta,i\mu}}{\partial v_{j\nu}} \delta \hat{v}_{j\nu} + \mathcal{O}(\delta \hat{v}^2). \quad (35)$$

The velocity-estimation error scales as $\delta \hat{v} \sim \Delta t^{1/2}$, so correlations containing two velocity errors are $\mathcal{O}(\Delta t)$. In contrast, $\delta \hat{v} \sim \Delta t^{-1/2}$ and therefore $\langle \delta \hat{v}_{i\mu} \delta \hat{v}_{j\nu} \rangle$ is $\mathcal{O}(\Delta t^0)$ and must be retained. More explicitly, Eq. (28) gives $\langle \delta \hat{v}_{i\mu} \delta \hat{v}_{j\nu} \rangle = \delta_{ij} \delta_{\mu\nu} \sigma^2 K_{av}$, whereas $\langle \delta \hat{v}_{i\mu} \delta \hat{v}_{j\nu} \rangle = \mathcal{O}(\Delta t)$.

We now consider the two terms appearing in Eq. (32) separately. Using $\hat{v}_{i\mu} = D_{i\mu} + \delta \hat{v}_{i\mu} + \mathcal{O}(\Delta t)$ where the true deterministic dynamics is denoted by $D_{i\mu}$, together with Eq. (34), we obtain

$$\begin{aligned} \left\langle \hat{v}_{i\mu} c_{i\mu,p}(\hat{\mathbf{v}}) \right\rangle &= \langle D_{i\mu} c_{i\mu,p}(\mathbf{v}) \rangle \\ &+ \sum_{j,\nu} \left\langle \delta \hat{v}_{i\mu} \delta \hat{v}_{j\nu} \frac{\partial c_{i\mu,p}}{\partial v_{j\nu}} \right\rangle + \mathcal{O}(\Delta t). \end{aligned} \quad (36)$$

At leading order, the derivative $\partial c_{i\mu,p} / \partial v_{j\nu}$ is evaluated at the underlying smooth state $\mathbf{v}$. Its dependence on the stochastic velocity-estimation error would introduce an additional factor of $\delta \hat{\mathbf{v}}$ and hence contributes only at higher order. The remaining stochastic average is therefore controlled by Eq. (28),

$$\begin{aligned} \sum_{j,\nu} \left\langle \frac{\partial c_{i\mu,p}}{\partial v_{j\nu}} \delta \hat{v}_{i\mu} \delta \hat{v}_{j\nu} \right\rangle &= \sigma^2 K_{av} \sum_{j,\nu} \left\langle \frac{\partial c_{i\mu,p}}{\partial v_{j\nu}} \right\rangle \delta_{ij} \delta_{\mu\nu} \\ &= \sigma^2 K_{av} \left\langle \frac{\partial c_{i\mu,p}}{\partial v_{i\mu}} \right\rangle. \end{aligned} \quad (37)$$

Thus, although $c_{i\mu,p}$ may depend on the velocities of other particles and on other Cartesian components, these derivatives do not contribute to the leading discretisation bias: the independence and isotropy of the dynamical noise make the acceleration-velocity cross-covariance diagonal in both particle and Cartesian indices.

$$\left\langle \hat{v}_{i\mu} c_{i\mu,p}(\hat{\mathbf{v}}) \right\rangle = \langle D_{i\mu} c_{i\mu,p}(\mathbf{v}) \rangle + K_{av} \sigma^2 \left\langle \frac{\partial c_{i\mu,p}}{\partial v_{i\mu}} \right\rangle + \mathcal{O}(\Delta t). \quad (38)$$

The second term in the loss gradient, $D_{\theta,i\mu}(\hat{\mathbf{v}})c_{i\mu,p}(\hat{\mathbf{v}})$, is also affected by velocity-estimation noise. However, substituting Eqs. (34) and (35), all terms linear in $\delta\hat{\mathbf{v}}$ have zero mean, while the first non-zero stochastic contribution contains two velocity errors and is therefore only $\mathcal{O}(\Delta t)$. Hence

$$\langle D_{\theta,i\mu}(\hat{\mathbf{v}})c_{i\mu,p}(\hat{\mathbf{v}}) \rangle = \langle D_{\theta,i\mu}(\mathbf{v})c_{i\mu,p}(\mathbf{v}) \rangle + \mathcal{O}(\Delta t). \quad (39)$$

Substituting Eqs. (38) and (39) into Eq. (32), and using $c_{i\mu,p} = \partial D_{\theta,i\mu} / \partial \theta_p$, gives

$$\begin{aligned} \frac{\partial \mathcal{L}_0}{\partial \theta_p} = & 2 \left\langle (D_{\theta,i\mu} - D_{i\mu}) \frac{\partial D_{\theta,i\mu}}{\partial \theta_p} \right\rangle \\ & - 2K_{av}\sigma^2 \left\langle \frac{\partial}{\partial v_{i\mu}} \left(\frac{\partial D_{\theta,i\mu}}{\partial \theta_p} \right) \right\rangle + \mathcal{O}(\Delta t). \end{aligned} \quad (40)$$

The first term is the desired regression gradient and the second is the discretisation bias. In particular, even if $\mathbf{D}_\theta = \mathbf{D}$, the second term need not vanish, so the true dynamics are not in general a fixed point of the naïve loss $\mathcal{L}_0$.

3. A loss with the leading bias removed

Equation (40) directly suggests how to remove the bias: add a term whose parameter derivative is equal and opposite to the second term above. Using the same averaging convention as the mean-squared loss, this gives

$$\mathcal{L}_{\text{corr}} = \left\langle \left(\hat{v}_{i\mu} - D_{\theta,i\mu} \right)^2 \right\rangle + 2K_{av}\sigma^2 \left\langle \frac{\partial D_{\theta,i\mu}}{\partial v_{i\mu}} \right\rangle. \quad (41)$$

Differentiating Eq. (41) with respect to any model parameter cancels the leading finite-difference contribution in Eq. (40), leaving

$$\frac{\partial \mathcal{L}_{\text{corr}}}{\partial \theta_p} = 2 \left\langle (D_{\theta,i\mu} - D_{i\mu}) \frac{\partial D_{\theta,i\mu}}{\partial \theta_p} \right\rangle + \mathcal{O}(\Delta t). \quad (42)$$

Thus $\mathbf{D}_\theta = \mathbf{D}$ is a fixed point to leading order in Δt . Importantly, this cancellation is performed directly at the level of the discretisation bias and therefore does not require $\langle \mathbf{v} \cdot \mathbf{F} \rangle = 0$ or conservative forces.

4. Correction for the model used here

Of the four contributions to the deterministic prediction,

$$\mathbf{D}_{\theta,i} = \mathbf{F}_i^{\text{env}} + \mathbf{F}_i^{\text{int}} + \mathbf{H}_i + \gamma'_{\text{eff},i} \cdot \mathbf{v}_i, \quad (43)$$

the two force terms depend only on positions, and $\mathbf{H}_i = -\tau \sum_{j \neq i} \mathbf{M}_{ij} \mathbf{v}_j$ depends on neighbour velocities but not on $\mathbf{v}_i$, so it contributes only to off-diagonal ($j \neq i$) entries and is annihilated by the δ_{ij} above. Since $\gamma'_{\text{eff},i}$ is itself a function of positions alone, the only velocity dependence is the explicit linear one, and the Jacobian is the drag matrix itself,

$$\frac{\partial D_{\theta,i\mu}}{\partial v_{i\nu}} = (\gamma'_{\text{eff},i})_{\mu\nu}, \quad \text{so} \quad \frac{\partial D_{\theta,i\mu}}{\partial v_{i\mu}} = (\gamma'_{\text{eff},i})_{\mu\mu} \quad (\text{no sum}). \quad (44)$$

Averaging Eq. (44) over the d Cartesian components, as Eq. (41) requires, therefore returns the normalised trace of the drag matrix,

$$\left\langle \frac{\partial D_{\theta,i\mu}}{\partial v_{i\mu}} \right\rangle_\mu = \frac{1}{d} \text{Tr } \gamma'_{\text{eff},i}. \quad (45)$$

D. Measurement error and the Savitzky–Golay filter

We model measurement error as an additive, zero-mean, temporally uncorrelated displacement of each tracked position, $x^{\text{obs}}[t] = x[t] + \eta[t]$ with $\langle \eta[t]\eta[t'] \rangle = \sigma_\eta^2 \delta_{tt'}$, independent of the dynamics. Under a naïve second difference this error is amplified by Δt^{-2} : the variance of $\hat{v}$ acquires a term $6\sigma_\eta^2/\Delta t^4$, which for realistic tracking accuracy can exceed the dynamical signal entirely.

Fitting a degree- d polynomial to a window of $2m + 1$ positions and differentiating it analytically suppresses this amplification, because the least-squares fit averages the independent errors over the window while remaining exact on the polynomial part of the trajectory [49]. We select (w, d) by requiring the autocorrelation of the residual $x^{\text{obs}} - x^{\text{sm}}$ to be minimal at all non-zero lags (Fig. 6b): a window that is too narrow leaves dynamical signal in the residual, while one that is too wide leaves correlated structure from over-smoothing. Once the residual is white, the remaining measurement-error contribution to $\langle \hat{v}\hat{v} \rangle$ is small. Fig. 6c–d shows that with this choice both the force field and τ are recovered across the noise levels tested, whereas the naïve estimator degrades rapidly.

E. Extension of the general-stencil correction to localization noise

The derivation above treats the observed positions as the true stochastic trajectory. Experimental trajectories additionally contain localization error. Unlike the intrinsic dynamical noise considered in Sec. VIC, localization noise enters not only the acceleration estimate but also the measured state on which the deterministic model is evaluated. Correcting only the induced acceleration-velocity covariance is therefore insufficient: the full localization-noise contribution to the model residual must be removed.

We write the observed position of particle i at frame n as

$$\mathbf{y}_{i,n} = \mathbf{x}_{i,n} + \boldsymbol{\epsilon}_{i,n}, \quad (46)$$

where $\mathbf{x}_{i,n}$ is the true position and $\boldsymbol{\epsilon}_{i,n}$ is the localization error. We assume zero-mean localization noise that is independent between frames, particles and Cartesian components,

$$\langle \epsilon_{i,n,\mu} \epsilon_{j,m,\nu} \rangle = \Lambda \delta_{ij} \delta_{\mu\nu} \delta_{nm}, \quad (47)$$

where Λ is the localization-error variance in one Cartesian component.

The position, velocity and acceleration entering the inference are all linear combinations of the measured positions. We write their localization-noise contributions on a common set of temporal offsets q as

$$\begin{aligned} \delta \hat{x}_{i\mu}^{\text{loc}} &= \sum_q c_q^x \epsilon_{i,q,\mu}, \\ \delta \hat{v}_{i\mu}^{\text{loc}} &= \frac{1}{\Delta t} \sum_q c_q^v \epsilon_{i,q,\mu}, \\ \delta \hat{a}_{i\mu}^{\text{loc}} &= \frac{1}{\Delta t^2} \sum_q c_q^a \epsilon_{i,q,\mu}. \end{aligned} \quad (48)$$

Here a coefficient is understood to be zero at offsets not used by the corresponding stencil. For S-G processing, c_q^x is the smoothing stencil, while c_q^v and c_q^a are the first- and second-derivative stencils. Any relative shift between the velocity and acceleration estimates is included in the offsets associated with these coefficients. For a scheme that evaluates the model at an unsmoothed central position, c_q^x reduces to a single unit coefficient at that frame.

Let

$$\hat{r}_{i\mu} = \hat{v}_{i\mu} - D_{\theta,i\mu}(\hat{\mathbf{x}}, \hat{\mathbf{v}}) \quad (49)$$

denote the residual entering the mean-squared loss. Expanding the deterministic model to first order in the localization error gives

$$\begin{aligned} \delta r_{i\mu}^{\text{loc}} &= \delta \hat{v}_{i\mu}^{\text{loc}} - \sum_{j,\nu} \frac{\partial D_{\theta,i\mu}}{\partial x_{j\nu}} \delta \hat{x}_{j\nu}^{\text{loc}} - \sum_{j,\nu} \frac{\partial D_{\theta,i\mu}}{\partial v_{j\nu}} \delta \hat{v}_{j\nu}^{\text{loc}} \\ &= \sum_{j,\nu,q} R_{i\mu,j\nu}^{(q)} \epsilon_{j,q,\nu}, \end{aligned} \quad (50)$$

where

$$R_{i\mu,j\nu}^{(q)} = \frac{c_q^a}{\Delta t^2} \delta_{ij} \delta_{\mu\nu} - c_q^x \frac{\partial D_{\theta,i\mu}}{\partial x_{j\nu}} - \frac{c_q^v}{\Delta t} \frac{\partial D_{\theta,i\mu}}{\partial v_{j\nu}}. \quad (51)$$

The derivatives of D_θ are evaluated at the measured local state. Equation (51) collects, in a single coefficient, every route by which a localization error at one position sample enters the residual: directly through the acceleration estimate, and indirectly through the position and velocity supplied to the model.

Using Eq. (47), the leading localization-noise contribution to the mean-squared residual is therefore

$$V_{\text{loc}}(\theta) = \Lambda \left\langle \sum_{j,\nu,q} \left(R_{i\mu,j\nu}^{(q)} \right)^2 \right\rangle_{i,\mu}. \quad (52)$$

This expression is exact for a deterministic model that is linear in the measured state and gives the leading correction for a smooth nonlinear model. Importantly, V_{loc} depends on the model parameters through the Jacobians of D_θ . It

must therefore be included in the training objective itself, rather than treated only as a constant contribution to the residual variance.

The distinction from the intrinsic dynamical-noise correction is important. For dynamical noise, the velocity-estimation error scales as $\delta \hat{v}^{\text{dyn}} \sim \Delta t^{1/2}$, so terms containing two such errors are $\mathcal{O}(\Delta t)$ and do not contribute to the leading bias derived in Sec. VIC. For localization noise,

$$\delta \hat{v}^{\text{loc}} \sim \frac{\sqrt{\Lambda}}{\Delta t}, \quad (53)$$

so quadratic predictor-error terms such as $\langle (\delta \hat{v}^{\text{loc}})^2 \rangle$ are not higher order and must be retained. This is why correcting only the localization-induced acceleration-velocity covariance does not in general remove the measurement-error bias.

Combining the two independent noise sources, the corrected loss becomes

$$\mathcal{L}_{\text{corr}}^{\text{loc}} = \left\langle \left(\hat{v}_{i\mu} - D_{\theta, i\mu} \right)^2 \right\rangle - V_{\text{loc}}(\theta) + 2K_{av}\sigma^2 \left\langle \frac{\partial D_{\theta, i\mu}}{\partial v_{i\mu}} \right\rangle. \quad (54)$$

The final term is the intrinsic dynamical-noise correction derived in Sec. VIC. Localization noise is now accounted for by subtracting its complete contribution $V_{\text{loc}}(\theta)$ from the observed residual variance.

The localization variance Λ is estimated from the raw, unfiltered position trajectories using localization-sensitive combinations of adjacent increments. The intrinsic noise amplitude σ^2 is estimated separately. When σ^2 is estimated from the model residual rather than supplied independently, the same localization contribution must first be removed:

$$\hat{\sigma}^2 = \frac{\Delta t}{K_{aa}} \left[\left\langle \left(\hat{v}_{i\mu} - D_{\theta, i\mu} \right)^2 \right\rangle - V_{\text{loc}}(\theta) \right], \quad (55)$$

to the same leading order used in the general-stencil derivation. In practice, residual-based estimation can additionally contain deterministic model error, so $\hat{\sigma}^2$ is most reliable once the fitted dynamics have converged. The localization correction itself is evaluated using the same masking and averaging convention as the mean-squared residual.

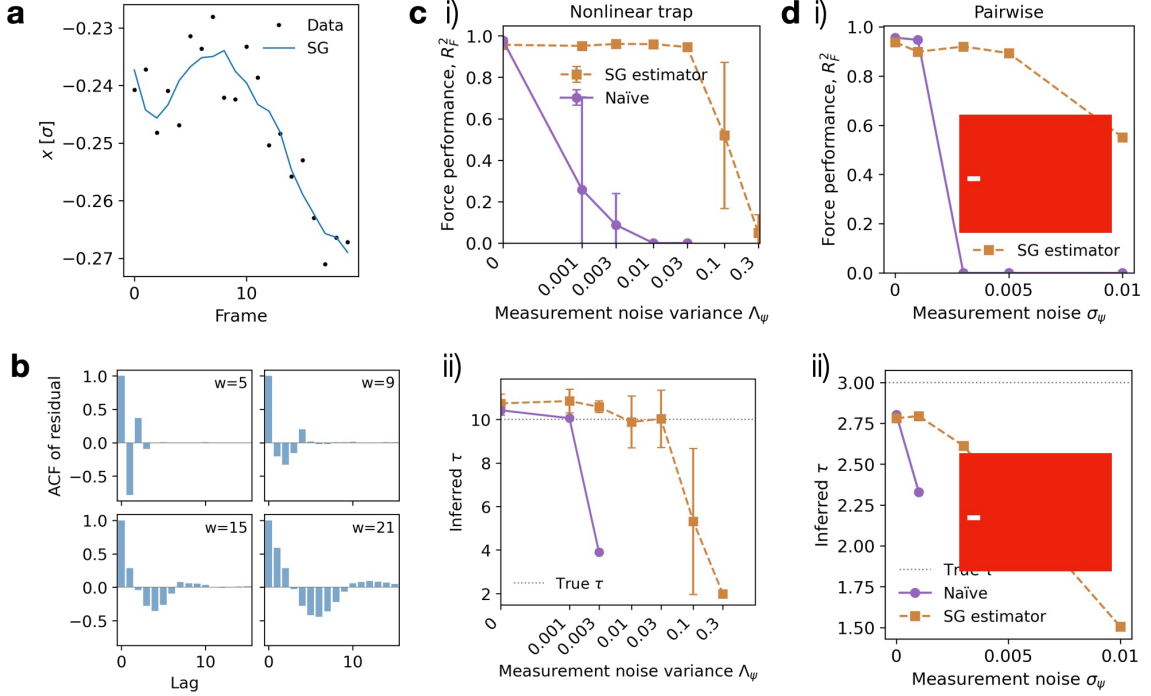


FIG. 6. **Robustness to measurement noise via Savitzky–Golay derivative estimation.** **a)** A sample simulated trajectory showing raw noisy positions (points) and the Savitzky-Golay (SG) smoothed trajectory (window $w = 9$, polynomial degree $p = 3$). **b)** Autocorrelation function (ACF) of the SG residual (raw – smoothed) averaged over all particles, for four candidate window sizes. A window whose residual ACF is minimal at all lags ≥ 1 is chosen as the most suitable filter. **c)** Performance of the inferred model for a nonlinear trap ($\alpha = 1.5$) as measurement noise σ_ψ is increased. i) Force performance $R_F^2 = 1 - \frac{\sum (F^T - F^I)^2}{\sum (F^T - \bar{F}^T)^2}$ and ii) inferred memory timescale τ , comparing the naïve finite-difference estimator (purple) with the SG derivative estimator and its associated Φ_{SG} correction (orange). The SG estimator maintains accurate force recovery and τ inference across all noise levels tested, while the naïve estimator degrades rapidly. **d)** **Pairwise not re-done with new scheme yet** Equivalent results for the pairwise soft-LJ system with periodic boundary conditions. The SG estimator (orange) is substantially more robust than the naïve scheme (purple).

VII. SI FIGURES

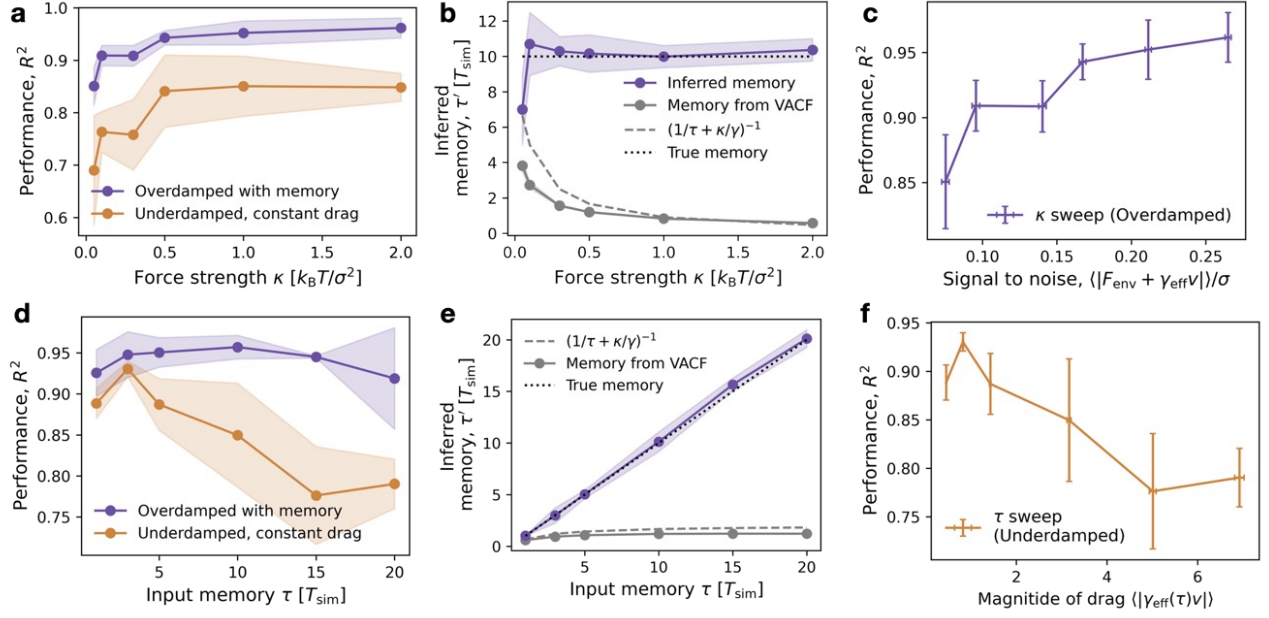


FIG. 7. **Needs new data from new loss Non-linear trends.** **a)** Performance R_F^2 (defined as in Fig. 2c) as the force strength κ is varied, comparing the overdamped inference scheme with memory (purple) to the naïve underdamped scheme with constant drag (orange). Shaded bands show ± 1 standard deviation across simulation replicates. **b)** Inferred memory timescale τ' for the same κ sweep, comparing the memory recovered by our scheme (purple), the memory estimated from the velocity autocorrelation function (VACF, grey), the effective-drag prediction $(1/\tau + \kappa/\gamma)^{-1}$ (dashed), and the true memory used to generate the trajectories (dotted). **c)** Performance R_F^2 from the same κ sweep as **a)**–**b)** overdamped case, re-expressed as a function of the signal-to-noise ratio $\langle |\mathbf{F}_{\text{env}} + \gamma_{\text{eff}} \mathbf{v}| \rangle / \sigma$. **d)** As in **a)**, but sweeping the input memory timescale τ instead of κ . **e)** As in **b)**, but for the τ sweep in **d)**. **f)** Performance R_F^2 from the same τ sweep as **d)**–**e)** underdamped case, re-expressed as a function of the magnitude of the effective drag, $\langle |\gamma_{\text{eff}}(\tau) \mathbf{v}| \rangle$. Simulation parameters and setup are detailed in the SI.

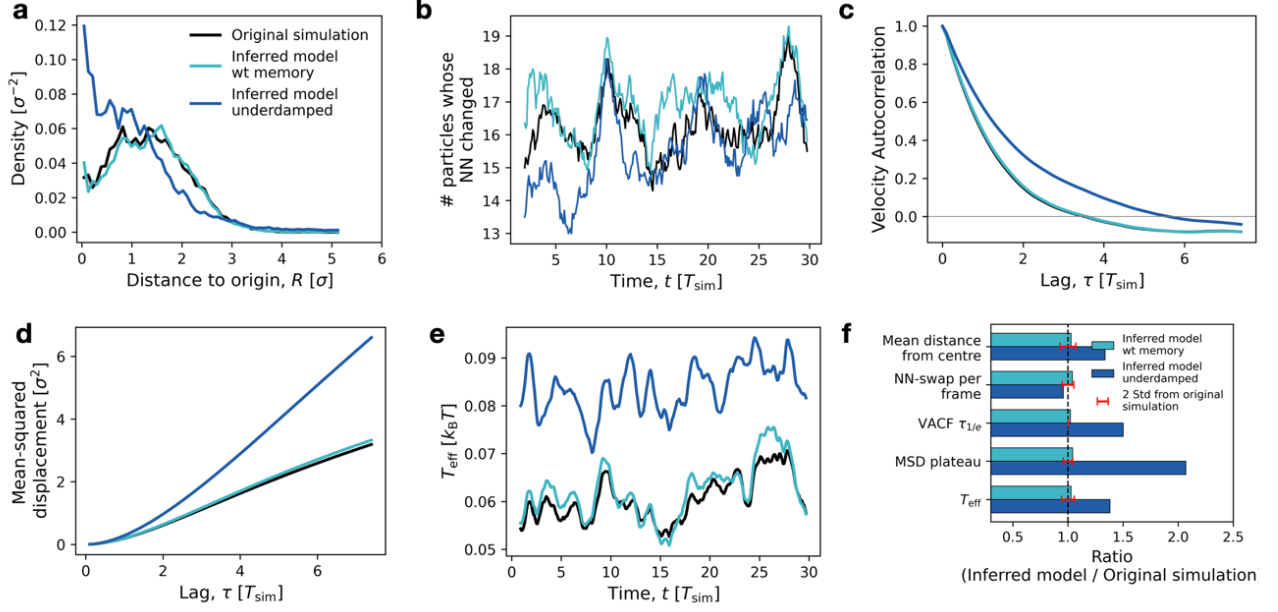


FIG. 8. **Statistical comparison of original and re-simulated trajectories using inferred force fields.** All simulations use $N = 100$ particles with parameters $\alpha = 1.5$, $\kappa = 0.5$, $\Gamma = 1$, $\tau = 10$, $\sigma = 1.1$, $T = 0.1$ (LJ units). All line colors correspond to the legend in (a). (a) Radial density profile: probability density of particle distance from the trap centre R [σ]. (b) Nearest-neighbour swap rate: number of particles whose identity of closest neighbour changes between consecutive frames, shown as a rolling average ($w = 10$ frames). (c) Normalised velocity autocorrelation function (VACF). (d) Mean-squared displacement (MSD) as a function of lag time τ . (e) Effective temperature measured from the displacements of particles per frame. (f) Summary of four key statistics expressed as the ratio of each re-simulated value to the original simulation value (dashed line at 1). Red error bars show the standard deviation across three independent seeds of the original simulation, indicating expected variability. The inferred model with memory (teal) reproduces the original statistics more faithfully than the underdamped model (blue) across all measures.

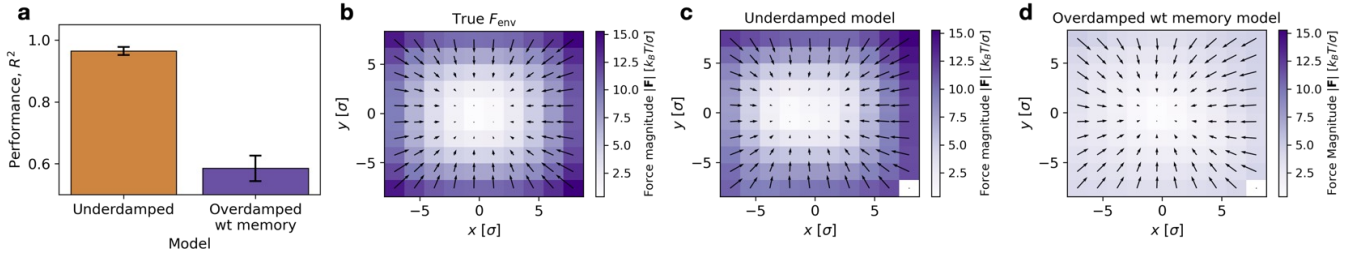


FIG. 9. **Needs new data from new loss Force inference from underdamped simulations.** (a) R^2 between the inferred and true environmental force field F_{env} for the underdamped model and the overdamped with memory model, evaluated across three random seeds (error bars: standard deviation). (b) True environmental force field $F_{\text{env}}(x, y)$ for a nonlinear harmonic trap ($\alpha = 1.5$). Colour indicates force magnitude; arrows show force direction. (c-d) Inferred force fields from the underdamped (c) and overdamped with memory (d) models, using the same colour scale as (b).

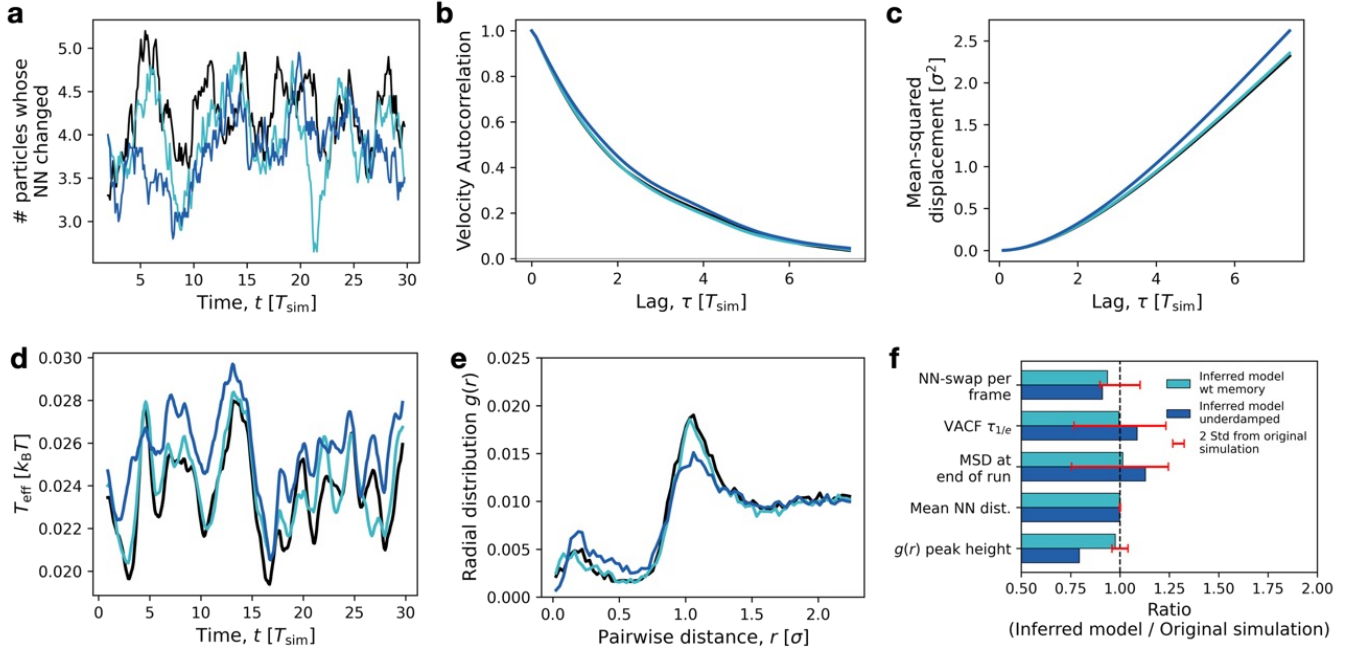


FIG. 10. **Statistical comparison of original and re-simulated trajectories using inferred force fields for pairwise setup.** All simulations use $N = 100$ particles with parameters $\epsilon = 0.3k_B T$, $\Gamma = 5$, $\tau = 5$, $\sigma = 1.1$, $T = 0.1$ (LJ units). All line colors correspond to the legend in (a). **(a)** Nearest-neighbour swap rate: number of particles whose identity of closest neighbour changes between consecutive frames, shown as a rolling average ($w = 10$ frames). **(b)** Normalised velocity autocorrelation function (VACF). **(c)** Mean-squared displacement (MSD) as a function of lag time τ . **(d)** Mean nearest neighbour distance. **(e)** Radial pair distribution function $g(r)$, measuring the likelihood of finding two particles separated by pairwise distance r relative to a uniform distribution. **(f)** Summary of the five key statistics expressed as the ratio of each re-simulated value to the original simulation value (dashed line at 1). Red error bars show two standard deviations across three independent seeds of the original simulation, indicating the expected variability. The inferred model with memory (teal) lies within the seed-to-seed range for all five statistics, whereas the underdamped model (blue) departs from it for the structural measures, most clearly the $g(r)$ peak height.

Appendix A: Extra derivation steps

1. Time derivative of the interaction term

Define the relative position and velocity as

$$\mathbf{r}_{ij} = \mathbf{x}_i - \mathbf{x}_j, \quad \Delta \mathbf{v}_{ij} = \dot{\mathbf{x}}_i - \dot{\mathbf{x}}_j, \quad (\text{A1})$$

so that $r_{ij} = |\mathbf{r}_{ij}|$ and $\hat{\mathbf{r}}_{ij} = \mathbf{r}_{ij}/r_{ij}$. The pairwise internal force is

$$\mathbf{f}_{ij}^{\text{int}} = f(r_{ij}) \hat{\mathbf{r}}_{ij}. \quad (\text{A2})$$

Taking the time derivative and applying the product rule,

$$\dot{\mathbf{f}}_{ij}^{\text{int}} = \dot{f} \hat{\mathbf{r}}_{ij} + f \dot{\hat{\mathbf{r}}}_{ij}. \quad (\text{A3})$$

The scalar $\dot{f} = (df/dr) \dot{r}_{ij}$, and $\dot{r}_{ij} = \hat{\mathbf{r}}_{ij} \cdot \Delta \mathbf{v}_{ij}$, giving

$$\dot{f} = \frac{df}{dr} (\hat{\mathbf{r}}_{ij} \cdot \Delta \mathbf{v}_{ij}). \quad (\text{A4})$$

Differentiating $\hat{\mathbf{r}}_{ij} = \mathbf{r}_{ij}/r_{ij}$ gives the transverse component of the relative velocity,

$$\dot{\hat{\mathbf{r}}}_{ij} = \frac{1}{r_{ij}} [\Delta \mathbf{v}_{ij} - \hat{\mathbf{r}}_{ij} (\hat{\mathbf{r}}_{ij} \cdot \Delta \mathbf{v}_{ij})]. \quad (\text{A5})$$

Combining,

$$\dot{\mathbf{f}}_{ij}^{\text{int}} = \frac{df}{dr} (\hat{\mathbf{r}}_{ij} \cdot \Delta \mathbf{v}_{ij}) \hat{\mathbf{r}}_{ij} + \frac{f}{r_{ij}} [\Delta \mathbf{v}_{ij} - \hat{\mathbf{r}}_{ij} (\hat{\mathbf{r}}_{ij} \cdot \Delta \mathbf{v}_{ij})]. \quad (\text{A6})$$

The first term is radial and the second is transverse to $\hat{\mathbf{r}}_{ij}$.

2. Velocity autocorrelation function

If we have no forces acting on our particles, and they obey :

$$\dot{x} = f^A/\gamma \quad (\text{A7})$$

$$\dot{f}^A = -\frac{1}{\tau} f^A + \xi \quad (\text{A8})$$

Then the velocity autocorrelation function should be proportional to the active force autocorrelation function: $\langle v(t)v(t+\Delta) \rangle = \frac{1}{\gamma^2} \langle f^A(t)f^A(t+\Delta) \rangle$. The solution for the active force in time takes the form:

$$f^A(t) = \int_{-\infty}^t e^{-(t-s)/\tau} \xi(s) ds \quad (\text{A9})$$

If we write out the autocorrelation of this function we get:

$$\langle f^A(t)f^A(t+\Delta) \rangle = \int_{-\infty}^t e^{-(t-s')/\tau} \xi(s') ds' \int_{-\infty}^t e^{-(t+\Delta-s)/\tau} \xi(s) ds \quad (\text{A10})$$

$$= \int_{-\infty}^t \int_{-\infty}^t e^{-(t-s')/\tau} e^{-(t+\Delta-s)/\tau} \sigma \delta(s-s') ds ds' \quad (\text{A11})$$

$$= \sigma \int_{-\infty}^t e^{-(2t-2s+\Delta)/\tau} ds \quad (\text{A12})$$

$$= \sigma e^{-\Delta/\tau} \int_{-\infty}^t e^{-2(t-s)/\tau} ds \quad (\text{A13})$$

$$= \frac{\sigma\tau}{2} e^{-\Delta/\tau} \quad (\text{A14})$$

So with no forces acting we should see a velocity autocorrelation $\langle v(t)v(t+\Delta) \rangle = \frac{\tau\sigma}{2\gamma^2} e^{-\Delta/\tau}$.

3. Correlations in discrete estimators

We find we need to use the backwards velocity and symmetric acceleration definitions for our inference scheme. Here we write out why we need these definitions. We write the discrete estimator for the acceleration of a particle at time t :

$$a^{\text{sym}} = \frac{1}{\Delta t^2} [x(t + \Delta t) - 2x(t) + x(t - \Delta t)] \quad (\text{A15})$$

$$= \frac{1}{\Delta t^2} [x_{n+1} - 2x_n + x_{n-1}] \quad (\text{A16})$$

where we have defined the n^{th} timestep as $t = n\Delta t$ and rewritten $x(n\Delta t + i\Delta t) = x_{n+i}$.

We can write three different versions for our velocity estimates:

$$v^{\text{fwd}} = \frac{1}{\Delta t} [x_{n+1} - x_n] \quad (\text{A17})$$

$$v^{\text{sym}} = \frac{1}{2\Delta t} [x_{n+1} - x_{n-1}] \quad (\text{A18})$$

$$v^{\text{bwd}} = \frac{1}{\Delta t} [x_n - x_{n-1}] \quad (\text{A19})$$

$$(\text{A20})$$

Now we define correlation functions between consecutive points x in our data (evolving under some unknown force fields).

$$C(0) = \sum_{n=0}^{n=N} x_n x_n \quad (\text{A21})$$

$$C(1) = \sum_{n=0}^{n=N} x_n x_{n+1} \quad (\text{A22})$$

$$C(2) = \sum_{n=0}^{n=N} x_n x_{n+2} \quad (\text{A23})$$

$$(\text{A24})$$

If true accelerations in our data depend in any way on the true velocities, we may go about inferring these relationships. This almost always boils down to calculating the correlation between our measured (discretized) acceleration and velocity:

$$\langle va \rangle = \sum_{n=0}^{n=N} v_n a_n \quad (\text{A25})$$

We can write this out for our three velocity estimates:

$$\langle v^{\text{fwd}} a^{\text{sym}} \rangle = \frac{1}{\Delta t^3} (3C(0) - 4C(1) + C(2)) \quad (\text{A26})$$

$$\langle v^{\text{sym}} a^{\text{sym}} \rangle = 0 \quad (\text{A27})$$

$$\langle v^{\text{bwd}} a^{\text{sym}} \rangle = \frac{1}{\Delta t^3} (-3C(0) + 4C(1) - C(2)) \quad (\text{A28})$$

where we have assumed time-translation invariance : $C(0) = \sum_{n=0}^{n=N} x_n x_n = \sum_{n=0}^{n=N} x_{n+1} x_{n+1} = \sum_{n=0}^{n=N} x_{n+i} x_{n+i}$. The only definition which gives the correct value for the inferred drag is (A28). Note that Eqs. (??) and (??) are equivalent to using the symmetric acceleration definition and backward velocity definition with a change of time variable $t \rightarrow +\Delta t$. This currently doesn't prove that it has to be backwards, just shows that both symmetric will give 0 correlations...

4. Model loss from maximum likelihood

We posit the generative model

$$\hat{v}_i = F' + \gamma'_* \hat{v}_i + \eta, \quad \eta \sim \mathcal{N}(0, A^2), \quad (\text{A29})$$

where $\eta \sim \mathcal{N}(0, A^2)$, taken independently and identically across particles and frames, has zero-mean, fixed variance A^2 , is temporally uncorrelated (white), and is independent of the trajectory data.

Equation (A29) specifies a probability distribution over the observed acceleration given the current local state. We denote by $\mathcal{C}_i = \{x_i, v_i, \{r_{ij}, v_j\}_{j \in N_i^{\text{neighs}}}\}$ the local configuration that cell i 's dynamics depend on, comprising its own position and velocity together with the relative positions and velocities of its neighbours $j \in N_i^{\text{neighs}}$. This is the full set of quantities entering the deterministic prediction $F' + \gamma'_* \hat{v}_i$. Because η is the only stochastic ingredient and is Gaussian, the model says that for a given local state $\mathcal{C}_i$ the observed $\hat{v}_i$ is itself Gaussian, centred on the deterministic prediction with variance A^2 :

$$p_\theta(\hat{v}_i | \mathcal{C}_i) = \frac{1}{\sqrt{2\pi A^2}} \exp\left(-\frac{[\hat{v}_i - (F' + \gamma'_* \hat{v}_i)]^2}{2A^2}\right). \quad (\text{A30})$$

This follows directly from Eq. (A29). Note that the conditioning state $\mathcal{C}_i$ enters only through the mean, variance A^2 does not depend on $\mathcal{C}_i$ and is constant.

Fitting the model by maximum likelihood means choosing the parameters (the MLP weights in F' , the scalar $\hat{\gamma}$ inside γ'_* , and the amplitude A , parametrised by θ) that make the observed data most probable under Eq. (A30). Since the samples are taken independently across particles and frames, the likelihood of the full dataset would equal the product over samples. We can work with the negative log-likelihood, which turns this product into a sum. For a single sample:

$$-\log p_\theta(\hat{v}_i | \mathcal{C}_i) = \frac{[\hat{v}_i - (F' + \gamma'_* \hat{v}_i)]^2}{2A^2} + \frac{1}{2} \log(2\pi A^2), \quad (\text{A31})$$

Maximising the likelihood for the full dataset is equivalent to minimising the sample-averaged negative log-likelihood, and this average will become our loss. We note that as the number of samples $N \rightarrow \infty$, the sample average converges to the expectation under the true data distribution $p_{\text{data}}(\hat{v}_i | \mathcal{C}_i)$:

$$\frac{1}{N} \sum_i [-\log p_\theta(\hat{v}_i | \mathcal{C}_i)]; \xrightarrow{N \rightarrow \infty} \langle -\log p_\theta(\hat{v} | \mathcal{C}) \rangle_{p_{\text{data}}} \quad (\text{A32})$$

which decomposes as

$$-\langle \log p_\theta \rangle_{p_{\text{data}}} = -\langle \log p_{\text{data}} \rangle_{p_{\text{data}}} + D_{\text{KL}}(p_{\text{data}} \| p_\theta). \quad (\text{A33})$$

Where $-\langle \log p_{\text{data}} \rangle_{p_{\text{data}}}$ is the entropy of the data and does not depend on the parameters in the model and $D_{\text{KL}}(p_{\text{data}} \| p_\theta) = \int p_{\text{data}} \log \frac{p_{\text{data}}}{p_\theta}$ is the relative entropy between the two distributions. Minimising (A32) with respect to our parameters is therefore equivalent to minimising the relative entropy $D_{\text{KL}}(p_{\text{data}} \| p_\theta)$. Our loss follows:

$$\mathcal{L}^\eta = -\frac{1}{N} \sum_i \log p_\theta(\hat{v}_i | \mathcal{C}_i) \quad (\text{A34})$$

$$= \frac{1}{2A^2} \langle [\hat{v}_i - (F' + \gamma'_* \hat{v}_i)]^2 \rangle + \frac{1}{2} \log(2\pi A^2). \quad (\text{A35})$$

Where the average is over all samples in the data.

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